

ASTHA KUMARI

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SUMMARY

Computer Science graduate (B.Tech CSE, 2026) with hands-on practical training in computer networking: LAN/WAN fundamentals, VLANs, subnetting, VPN routing and the OSPF, EIGRP and RIP routing protocols, gained through the Advanced Certificate Course in HPC-AI at C-DAC Mohali. Working as a Technical Intern at CDAC Mohali with strong Linux and troubleshooting exposure, plus a Microsoft Azure Internship (AICTE–Microsoft Elevate Program). Keen to monitor, maintain and troubleshoot enterprise network infrastructure with clear verbal and written communication.

SKILLS

Networking: LAN/WAN Fundamentals, Subnetting, VLAN, VPN Routing, Network Troubleshooting

Routing Protocols: OSPF, EIGRP, RIP

Linux & Scripting: Linux, Linux Commands, Bash Scripting, Python

Programming Languages: Python, C, C++

HPC & Parallel Computing: OpenMP, MPI (Collective Communication), Parallel File Systems, Omni-Path Architecture

Cloud & BI: Microsoft Azure, Resource Groups, Azure Storage Account, Azure Cosmos DB, Azure AI Services, Azure Machine Learning, Power BI

AI & Machine Learning: Supervised and Unsupervised Learning, CNN, Computer Vision

Development & Database: JavaScript, React.js, HTML, CSS, Mern stack, SQL, DBMS

Core Subjects: Computer Networks, OS, DSA, SDLC, Cloud Computing, OOP's, DBMS

PROFESSIONAL EXPERIENCE

Advanced Certificate Course in HPC-AI (C-HUK Project)

Jul 2026 - Present

CDAC Mohali (Centre for Development of Advanced Computing), Punjab, India

- Completed hands-on networking practicals on **OSPF, RIP and EIGRP** routing protocols, **VLANs, subnetting and VPN routing**
- Studied **Computer Networking Fundamentals** covering LAN/WAN concepts, addressing and routing
- Worked on **Linux administration, Linux commands and Bash scripting** labs, plus **Python programming fundamentals**
- Learning **parallel programming** with **OpenMP** and **MPI collective communication**, and HPC infrastructure including **Omni-Path architecture** and **parallel file systems**

Technical Intern – Cybersecurity & AI/ML

Feb 2026 - Jul 2026

CDAC Mohali (Centre for Development of Advanced Computing), Punjab, India

- Gaining hands-on exposure to **Linux environments**, system security concepts, and secure software development practices
- Working on **cybersecurity and AI/ML-based projects** focused on secure system design and intelligent threat analysis
- Assisting in the development and testing of **machine learning models** for data analysis and pattern recognition
- Developed **DeepGuard AI**, an AI-powered Deepfake Detection System using **PyTorch, OpenCV, ResNet18, and Bidirectional LSTM** for identifying manipulated videos.

Microsoft Azure Intern (AICTE – Microsoft Elevate Program)

Jan 2026

Microsoft (Remote)

- Completed **Microsoft Azure Internship** under the **AICTE – Microsoft Elevate Program**
- Worked hands-on with **Azure Resource Groups, Storage Accounts, Azure Cosmos DB, Azure AI Services, and Machine Learning Workspace**
- Created and managed cloud resources using the **Azure Portal**
- Gained practical experience in **cloud administration, AI-based services, and resource management**
- Developed foundational understanding of **scalable cloud infrastructure and Azure service integration**

PROJECTS

Network Routing & Switching Practicals | C-DAC Mohali (HPC-AI Course)

- Practiced dynamic routing configuration using **RIP, OSPF and EIGRP** across multiple routers.
- Configured **VLANs** for LAN segmentation and worked on **subnetting** and IP address planning.
- Implemented **VPN routing** and diagnosed connectivity issues to resolve network problems.
- **Concepts:** LAN/WAN, Routing, Switching, VLAN, Subnetting, VPN, OSPF, EIGRP, RIP

Real-Time Sign Language Translator (Sign Lang AI) | [Link](#)

- Developed an AI-powered real-time sign language translation system to bridge communication gaps for deaf and mute individuals.
- Implemented live webcam hand gesture recognition with instant text generation and speech output, supporting A–Z alphabet and 33+ sign gestures.
- **Technologies:** Python, Streamlit, MediaPipe, OpenCV, TensorFlow, NumPy, Pandas, Scikit-Learn.

Class Vision-AI | [Link](#)

- Designed and deployed a **cloud-based AI classroom analytics platform** on Microsoft Azure.
- Achieved **real-time face recognition and attendance automation** with a Streamlit **web dashboard**; deployed using **Azure Cloud services and GitHub integration**.
- **Technologies:** Python, OpenCV, Deep Learning, NumPy, Machine Learning, Streamlit, Azure, GitHub

DeepGuard AI – Deepfake Detection System | [Link](#)

- Developed an **AI-powered Deepfake Detection Web Application** using **ResNet18 and Bi-LSTM architecture** for temporal video analysis and REAL/FAKE prediction with confidence scoring.
- Implemented a pipeline of **Frame Extraction, Face Detection, Face Alignment, Feature Extraction and Classification**, with user authentication and analysis history.
- **Technologies:** Python, PyTorch, OpenCV, Streamlit, ResNet18, Bi-LSTM, NumPy, Pandas

EDUCATION & CERTIFICATIONS

I.K.G. PTU Mohali – 1, Punjab, India (2022 – 2026)

Bachelor of Technology (B. Tech) in CSE

CGPA: 8.6/10

- Advanced Certificate Course in HPC-AI – CDAC Mohali (C-HUK Project), Jul 2026 - Present
- Microsoft Azure Internship Certificate (AICTE – Microsoft Elevate, Jan 2026) | [Link](#)
- Course Completion Certificates: Power BI for Business Applications | [Link](#)
- Course Completion Certificates: Microsoft Azure | [Link](#)
- Course Completion Certificates: Artificial Intelligence & Machine Learning | [Link](#)
- Course Completion Certificates: Cloud Administration & Engineering | [Link](#)
- 2b Academy Cybersecurity Career Workshop | [Link](#)